Programming: Analysing Ocean Drifters

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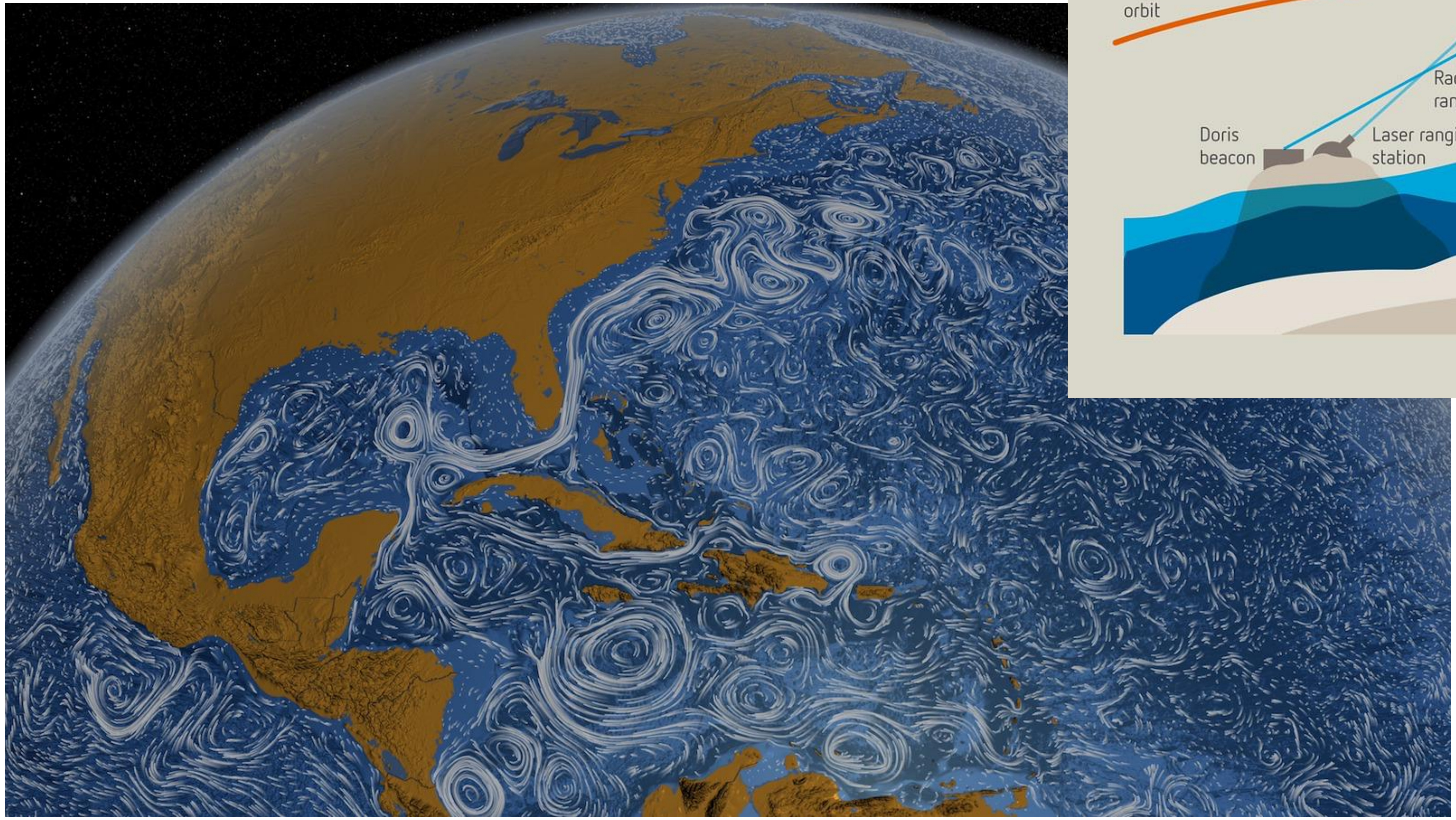
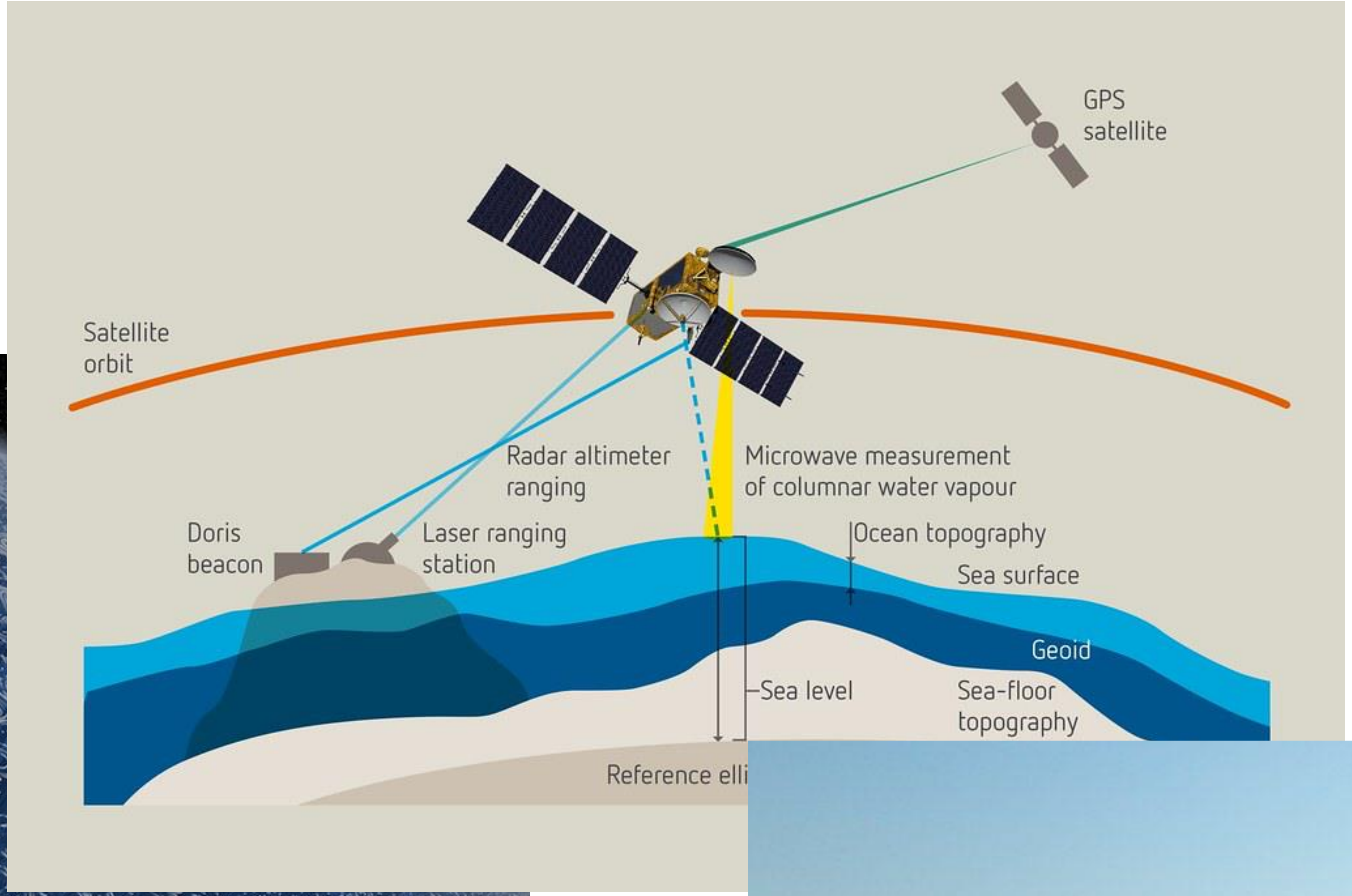
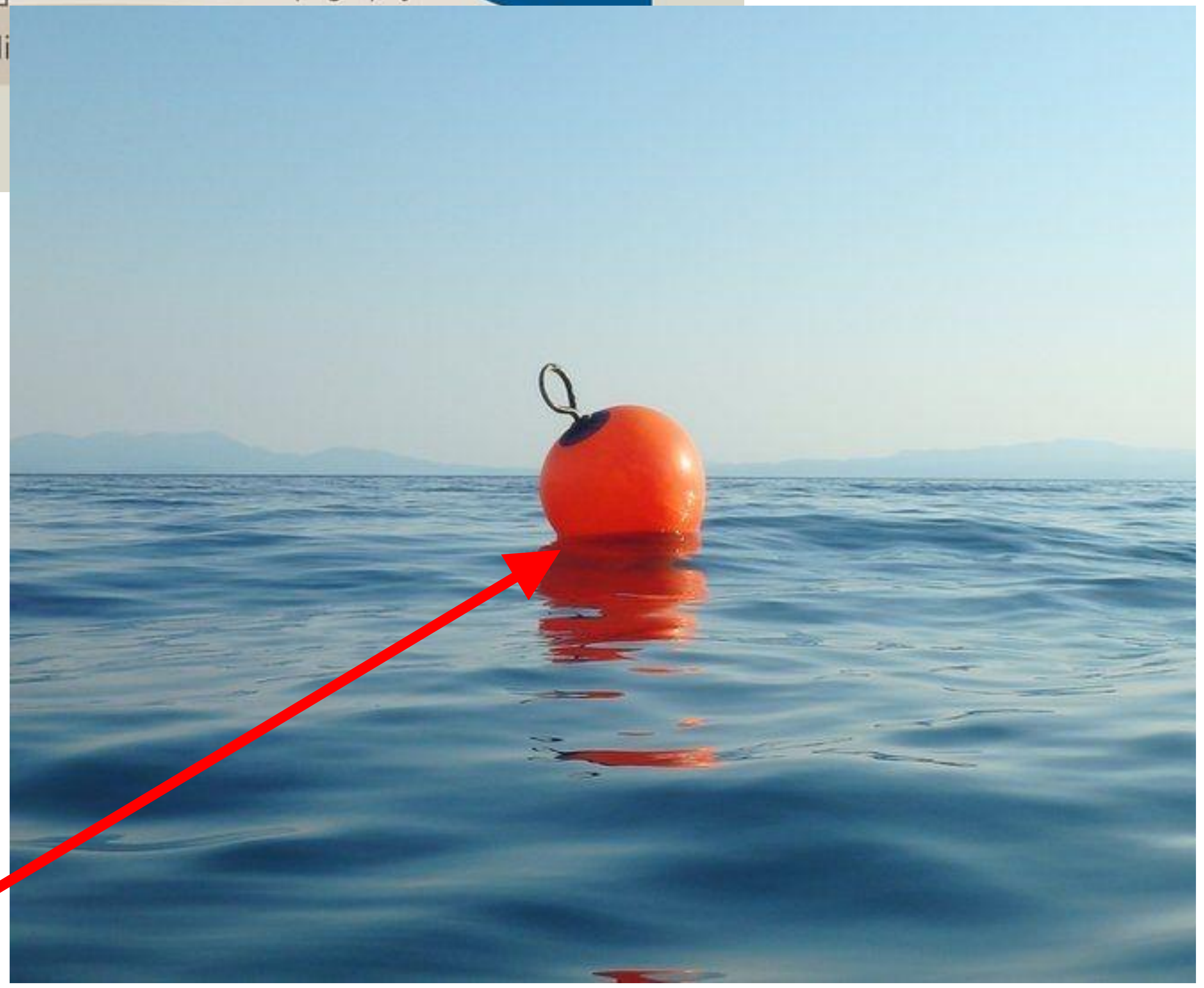
The goal(s) of this session

- Learning about ocean drifter data, what it is and how to model and analyse it
- Learning how to do statistical analysis and visualisations in R (or Python)
- Bonus: working in small groups, learning to communicate, code efficiently and use resources!
- This Wednesday (9.30am), prepare one favourite visualisation from today's programming session! Email it to me by next Tuesday evening (adam.sykulski@imperial.ac.uk)

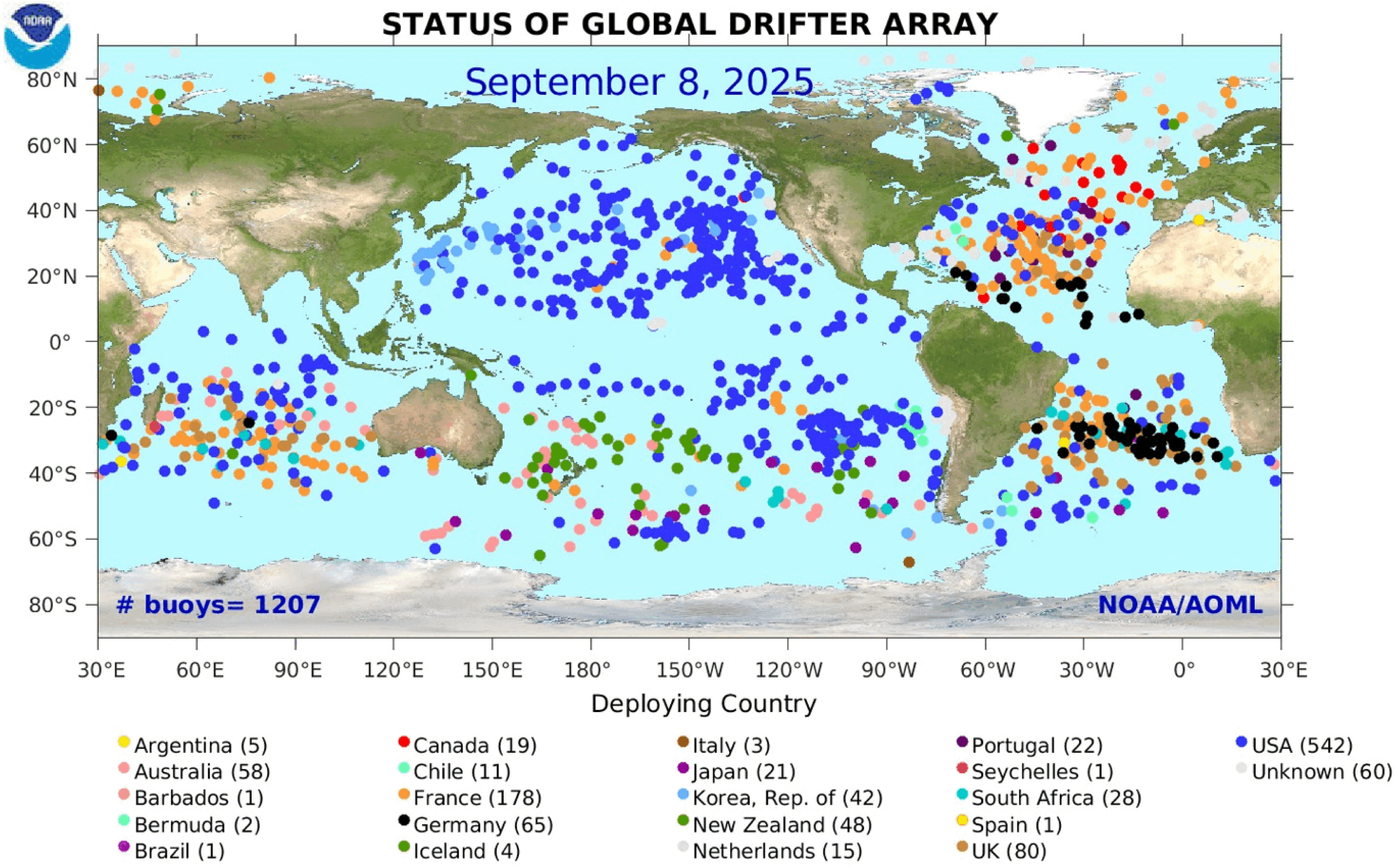
Monitoring our Oceans





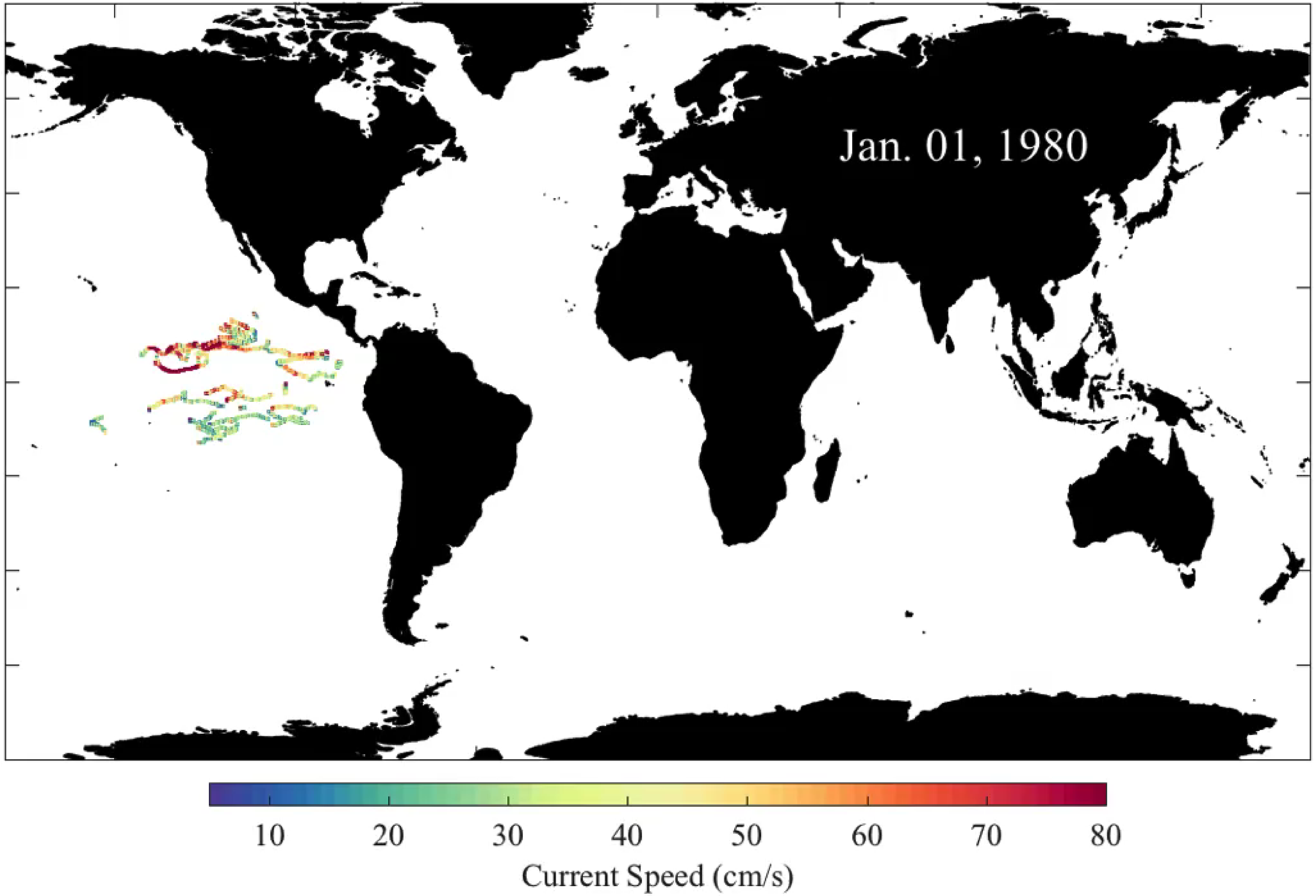


“Buoy” or “Drifter”

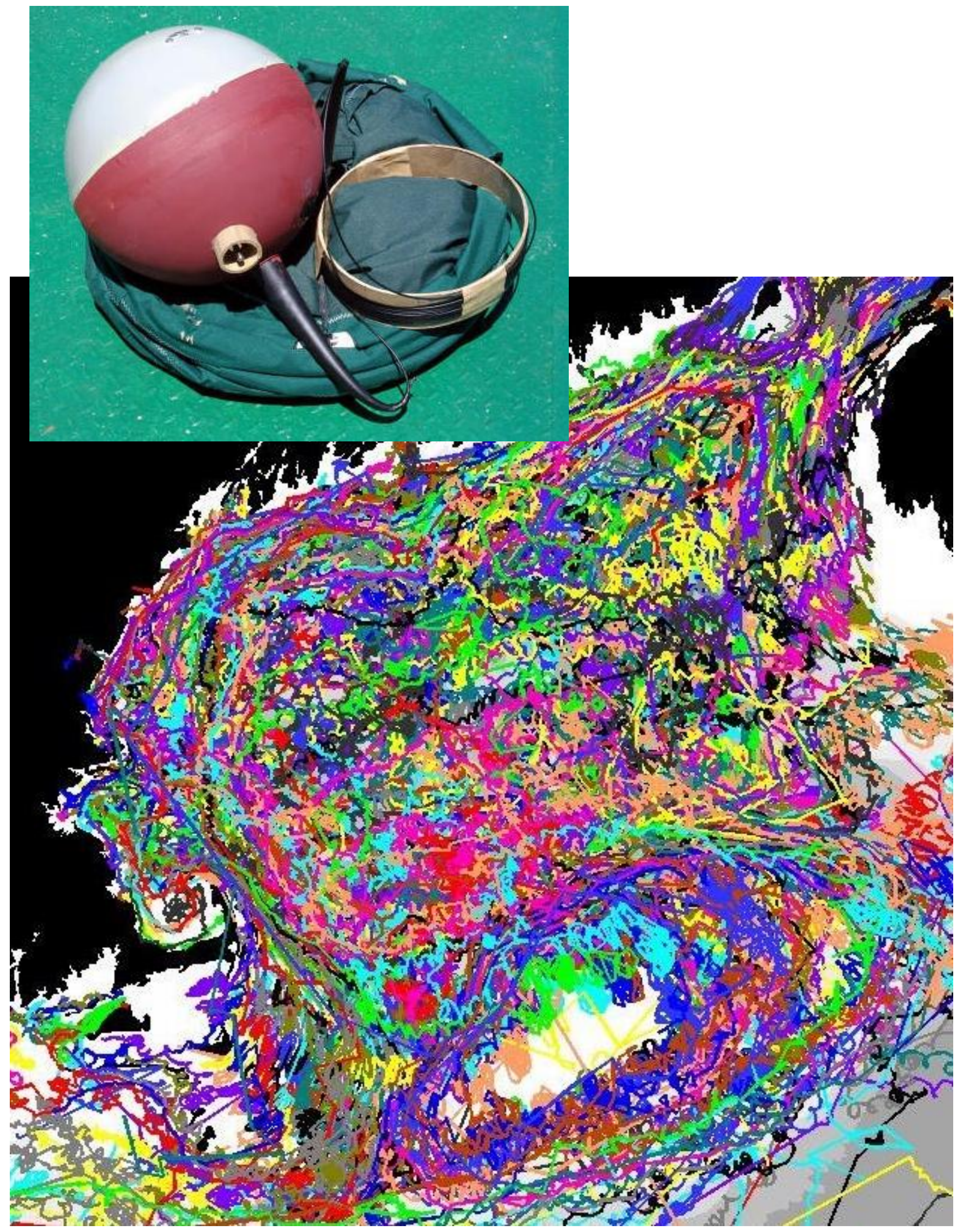


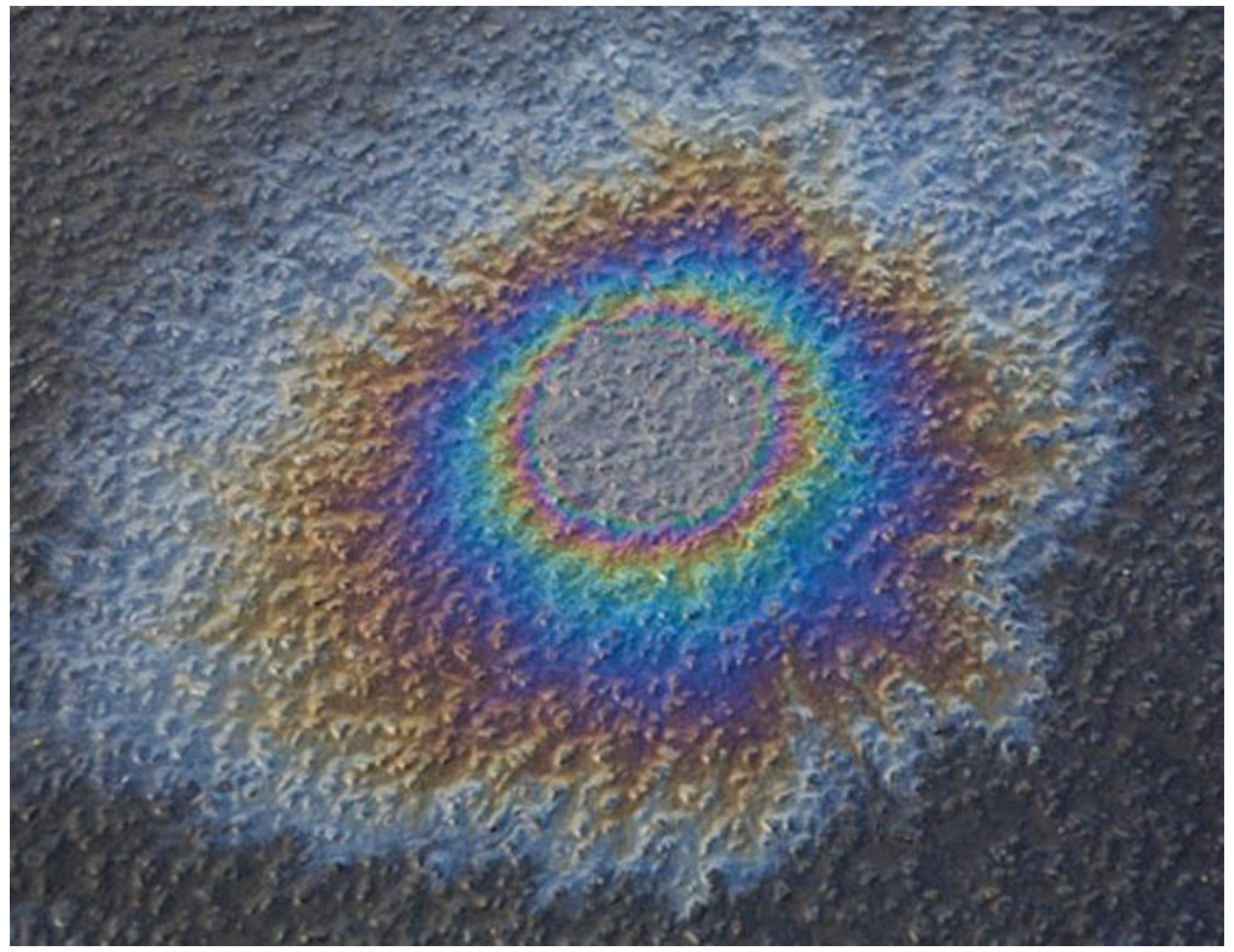
STATUS OF GLOBAL DRIFTER ARRAY
September 8, 2025
80°N
60°N
40°N
20°N
0°
20°S
40°S
60°S
80°S
buoys= 1207
NOAA/AOML
30°E
60°E
90°E
120°E
150°E
180°
150°W
120°W
90°W
60°W
30°W
0°
30°E
Deploying Country
Argentina (5)
Australia (58)
Barbados (1)
Bermuda (2)
Brazil (1)
Canada (19)
Chile (11)
France (178)
Germany (65)
Iceland (4)
Italy (3)
Japan (21)
Korea, Rep. of (42)
New Zealand (48)
Netherlands (15)
Portugal (22)
Seychelles (1)
South Africa (28)
Spain (1)
UK (80)
USA (542)
Unknown (60)

Global Drifter Program

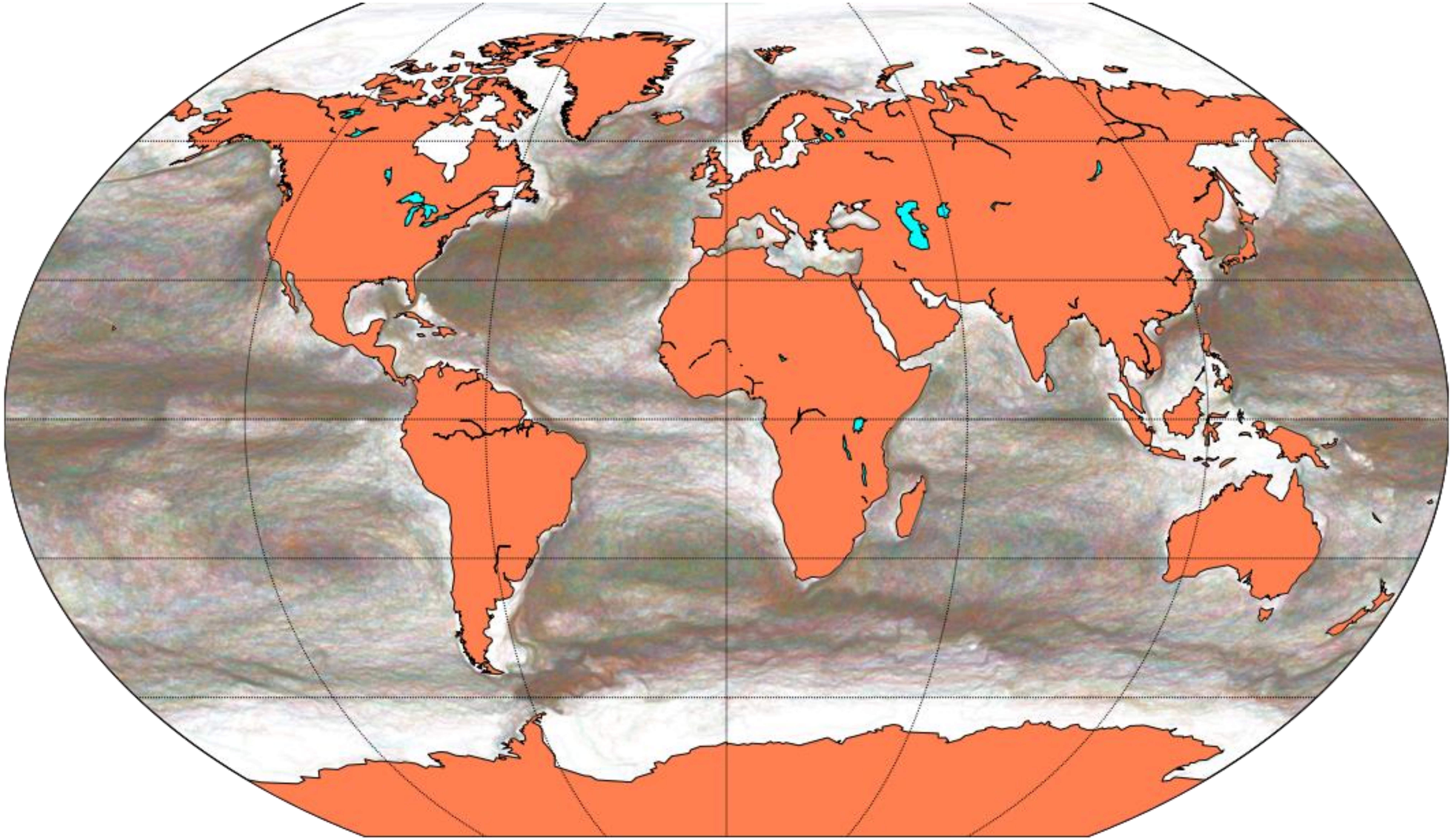


Wait…what is this good for?

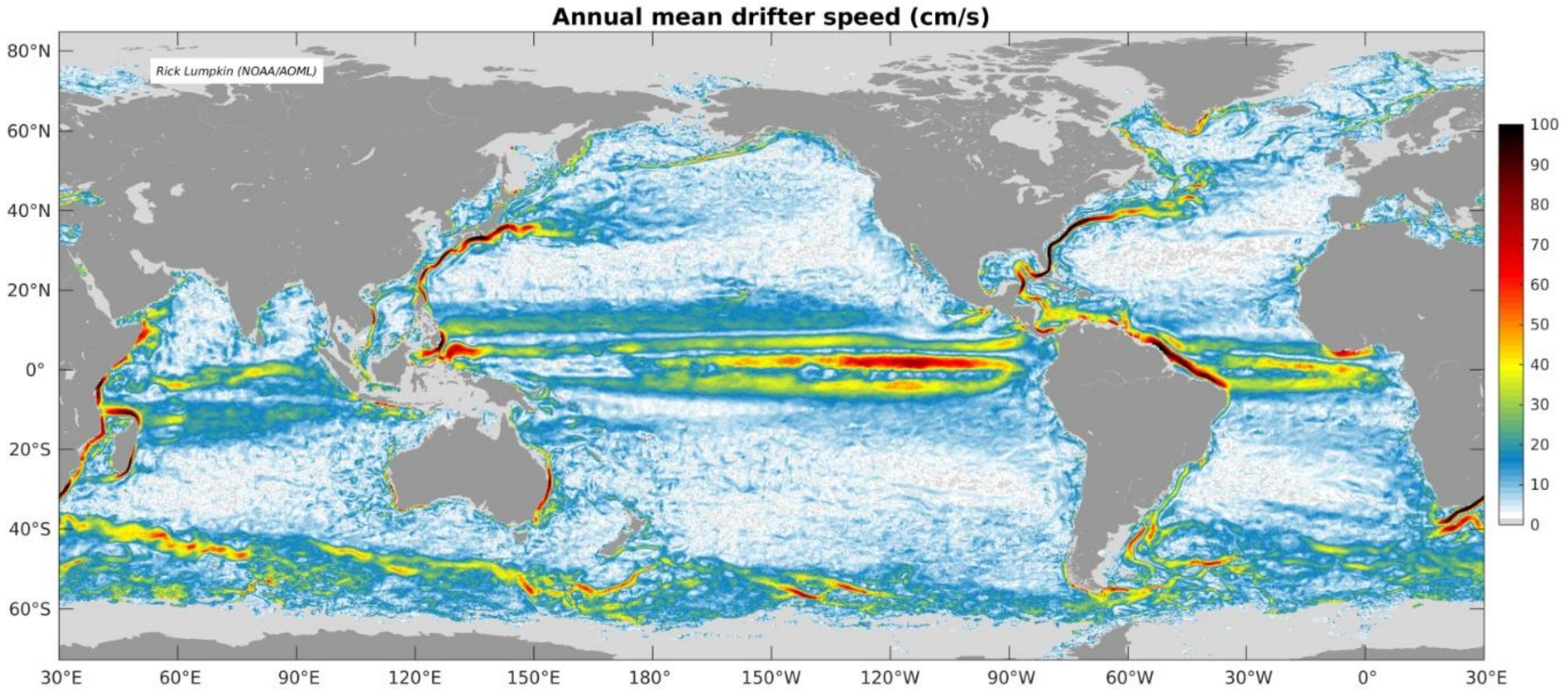




Spaghetti Plot – “data dump”

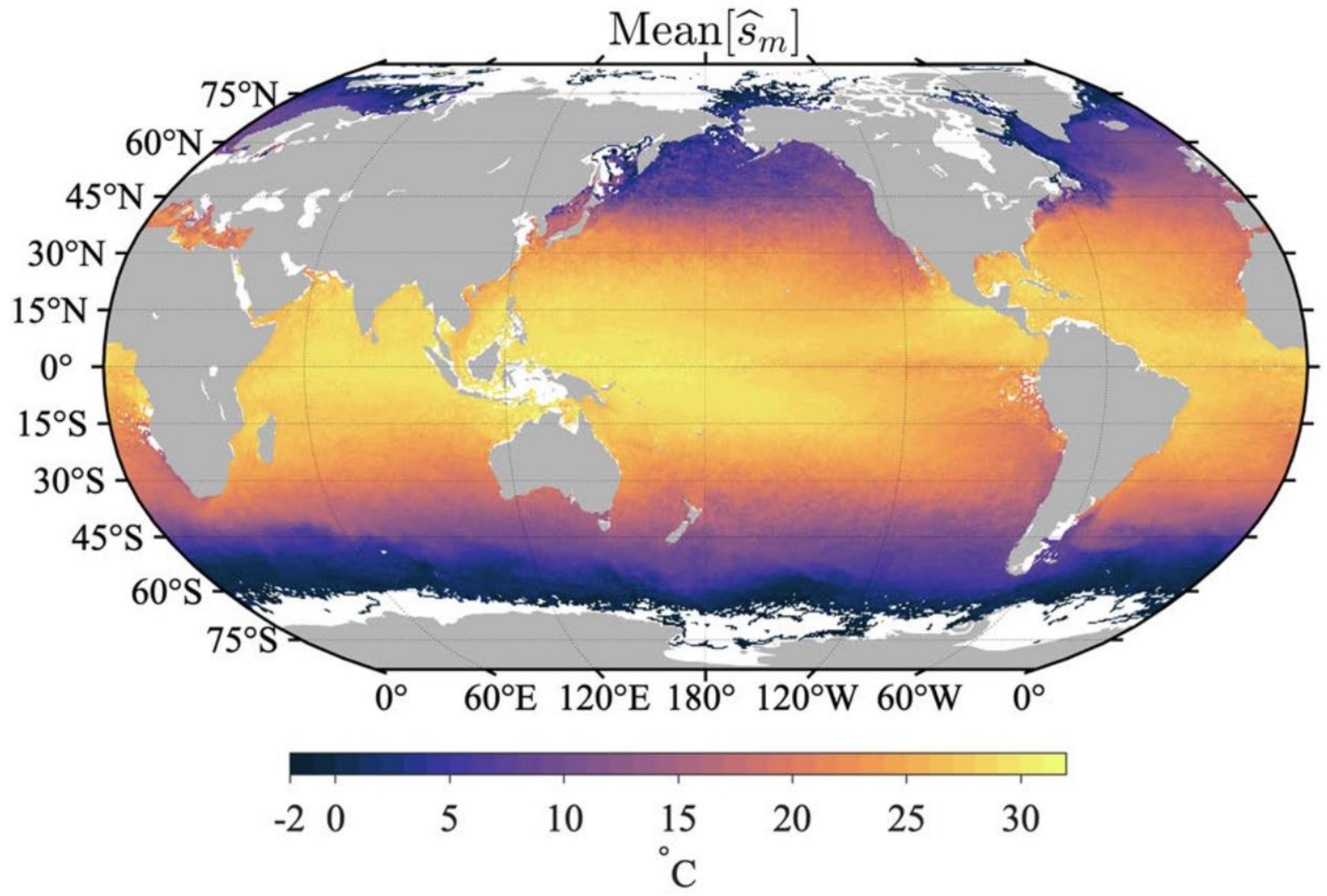


First step: Averaging the data

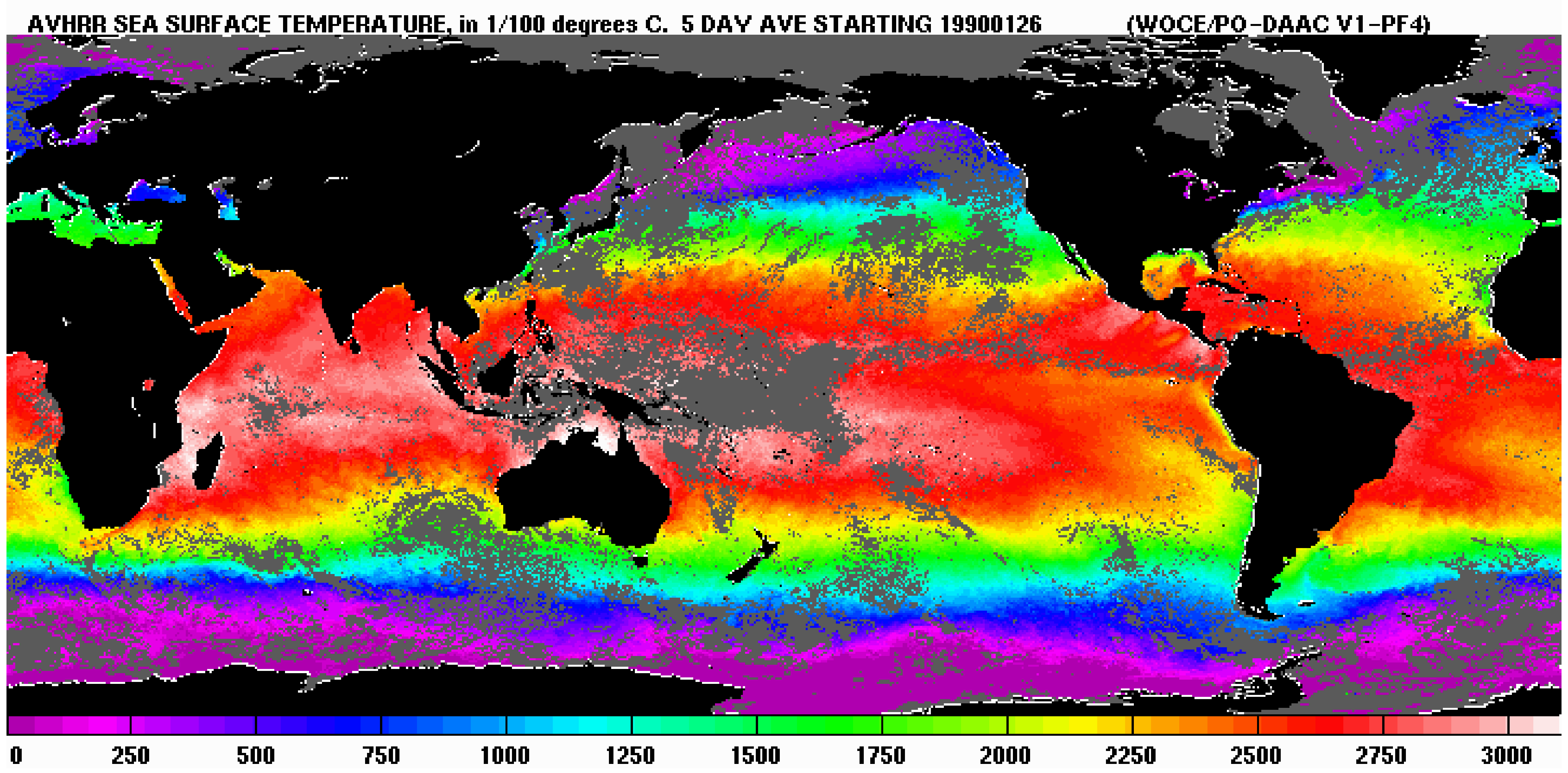


Activity 1: Drifters vs Satellites

Sea Surface Temperature



Drifter SST



Satellite SST

Download data and R scripts from:
github.com/AdamSykulski/OceanDrifterDataInR.git

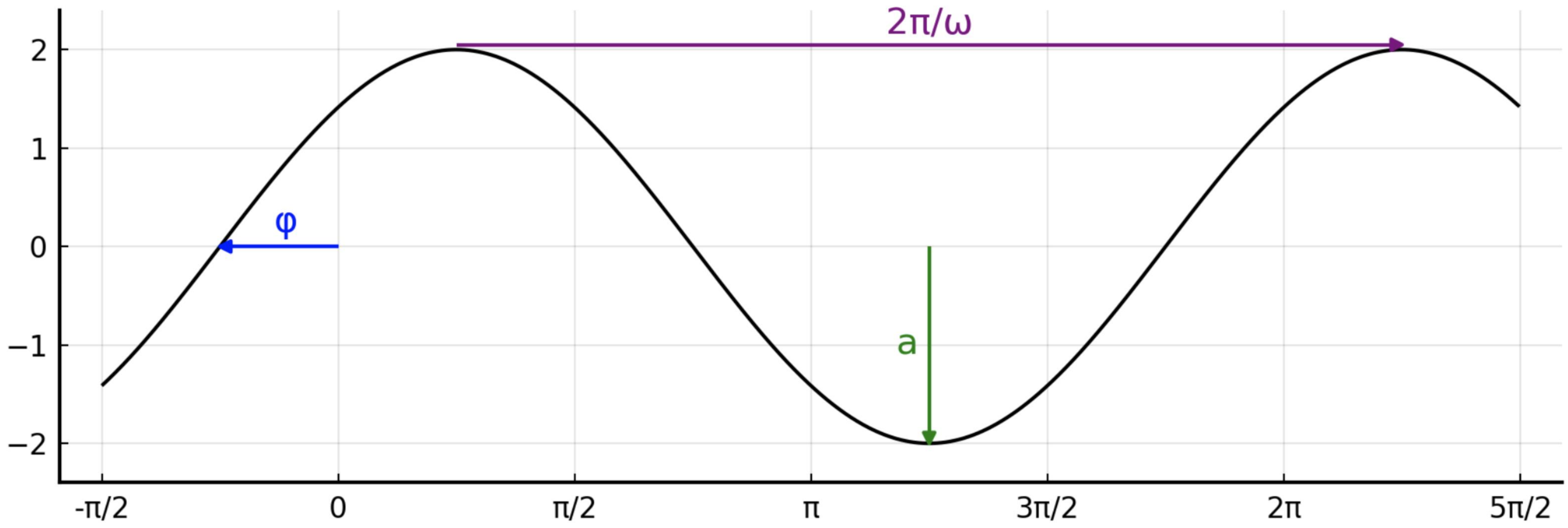
And then head over to RStudio for Activity 1!

Activity 2: Spectral Analysis of Time Series

Consider a basic sinusoid:

$$\eta(t) = a\sin(\omega t + \varphi)$$

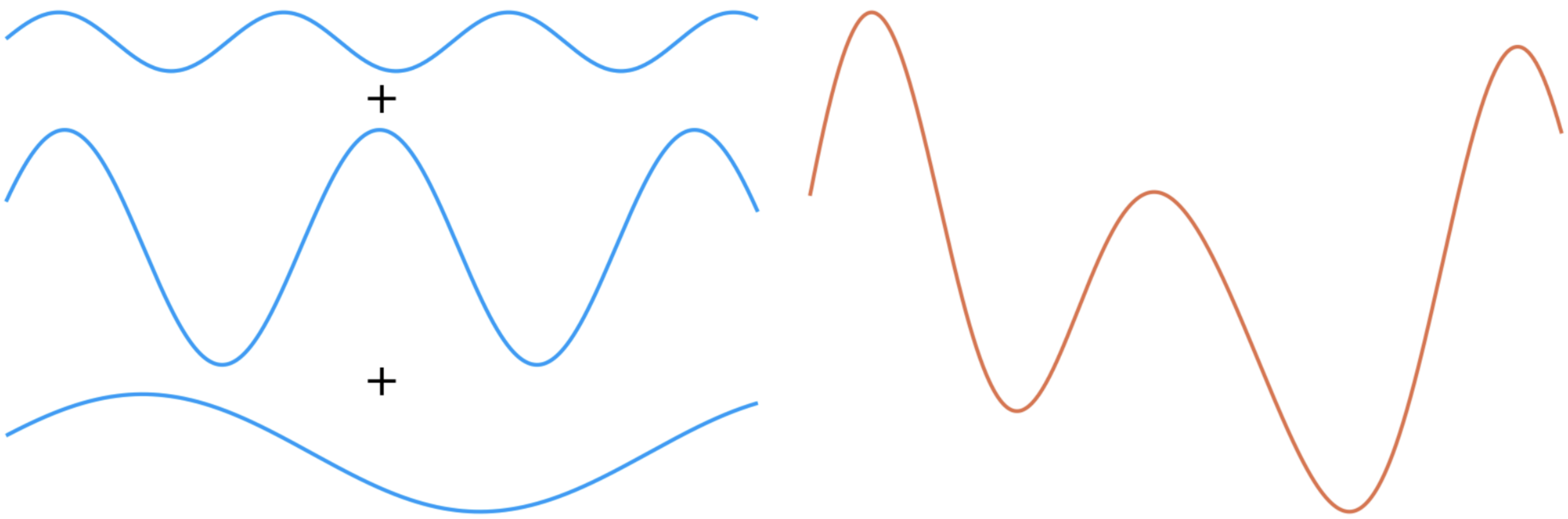
- a is the *amplitude*.
- φ is the *phase*.
- ω is the *angular frequency*.



We can represent more complicated functions as a sum of sinusoids:

$$\eta(t) = \sum_{\omega} a(\omega) \sin(\omega t + \varphi(\omega))$$

with different amplitudes and phases for each component

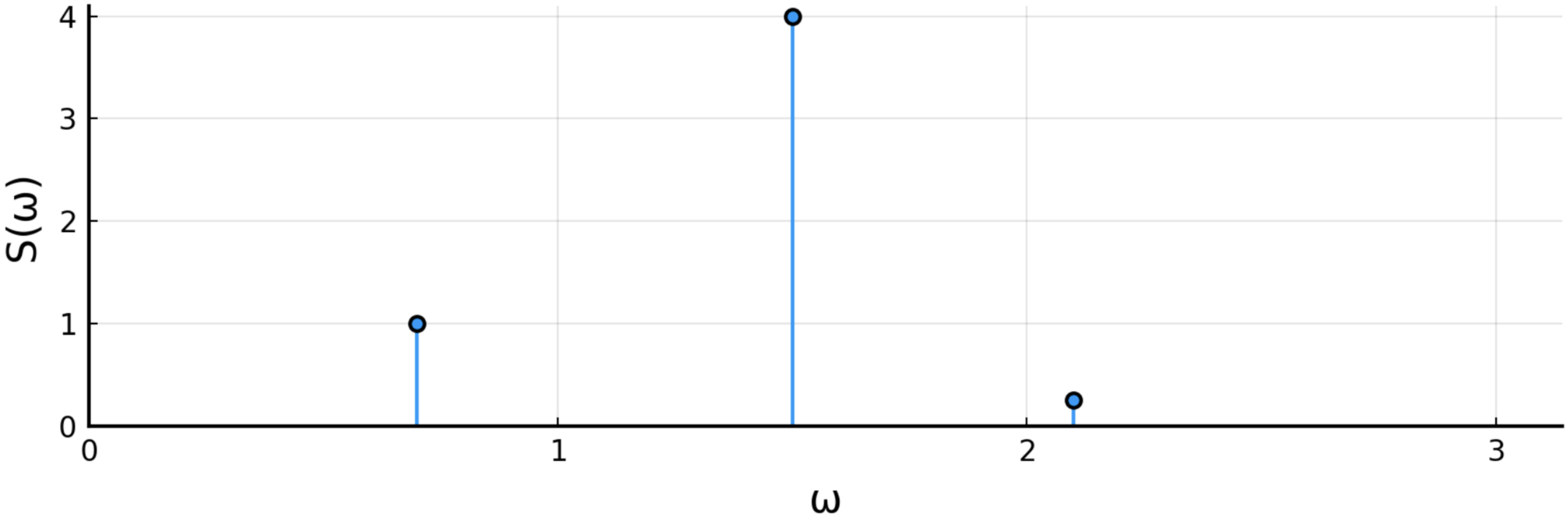


The power spectrum: discrete frequency

The power spectrum is then defined as

$$S(\omega) = |a(\omega)|^2$$

In our example:



We could keep going … but what if our time series process lives at all frequencies and is stochastic?

This is the field of spectral analysis!

Estimating the power spectral density

Theory: $S_x(\omega) = \lim_{T\to\infty} \mathbb{E}\left(\frac{1}{2T} \left| \int_{-T}^{T} x(t) e^{-i\omega t} dt \right|^2 \right)$

Practice: Observe some sample $X_1, \ldots, X_N$ at intervals Δ such that

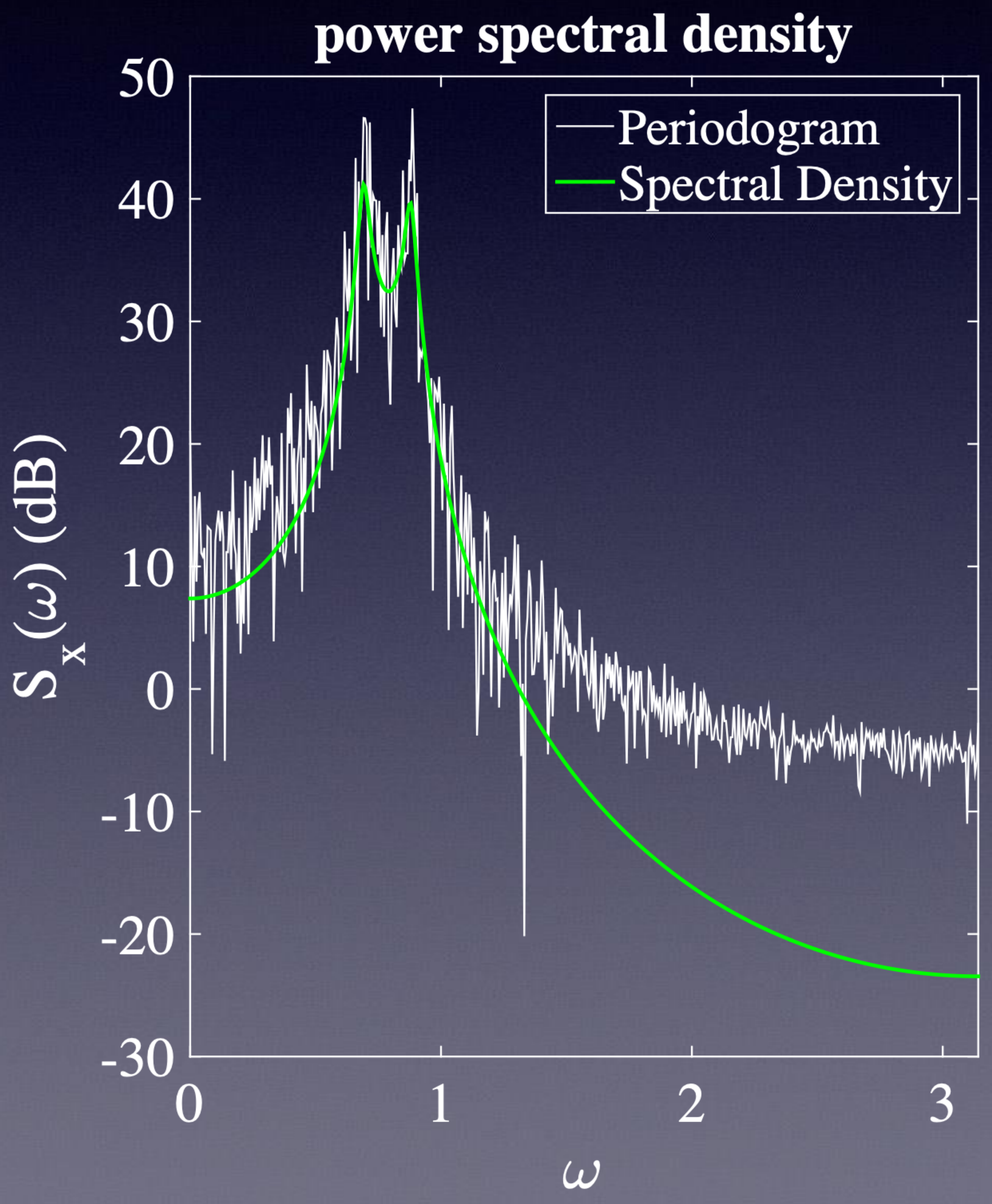
$$\hat{S}(\omega) = \frac{\Delta}{N} \left| \sum_{t=1}^{N} X_t e^{-i\omega t} \right|^2$$

This is called the *periodogram* and is defined for $\omega \in [-\pi/\Delta, \pi/\Delta]$ where π/Δ is the *Nyquist frequency*.

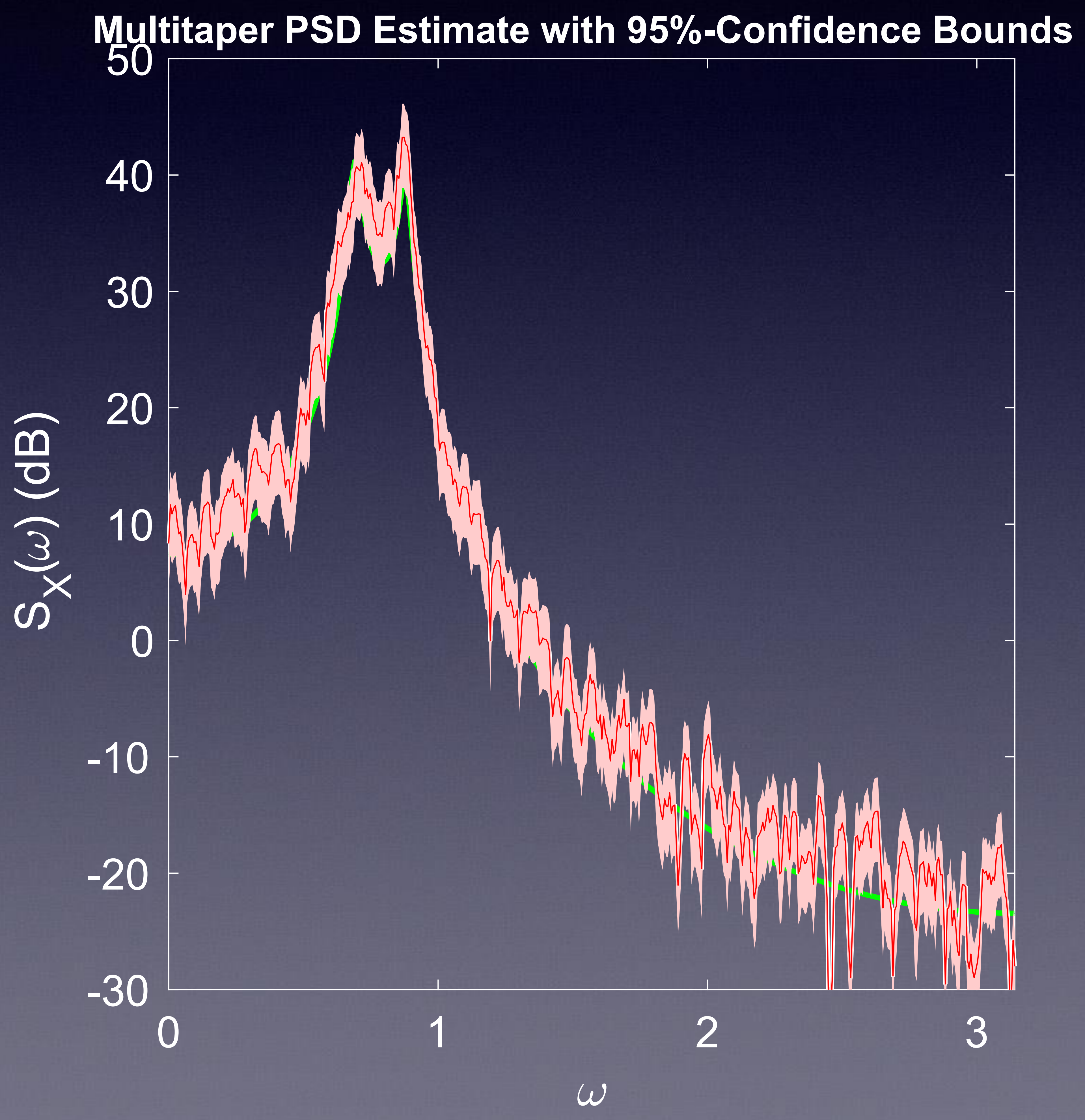
AR(4)

$$x_t = 2.7607x_{t-1} - 3.8106x_{t-2} + 2.6535x_{t-3} - 0.9238x_{t-4} + \varepsilon_t,\ \varepsilon_t \sim \mathcal{N}(0,1)$$

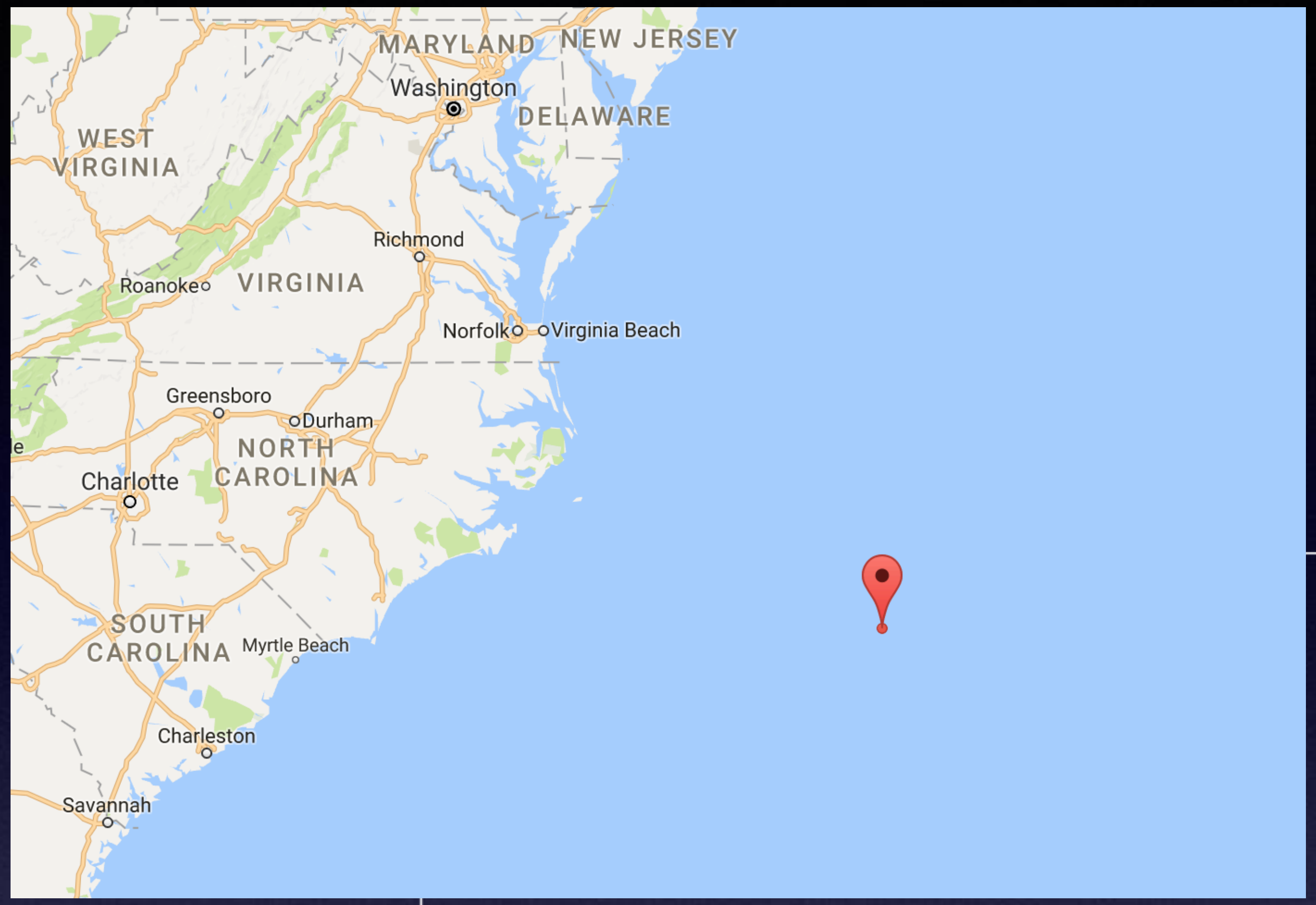
$$S_x(\omega) = \frac{\sigma_\epsilon^2}{|1 - \sum_{k=1}^{4} \phi_k e^{-ikw}|^2}$$

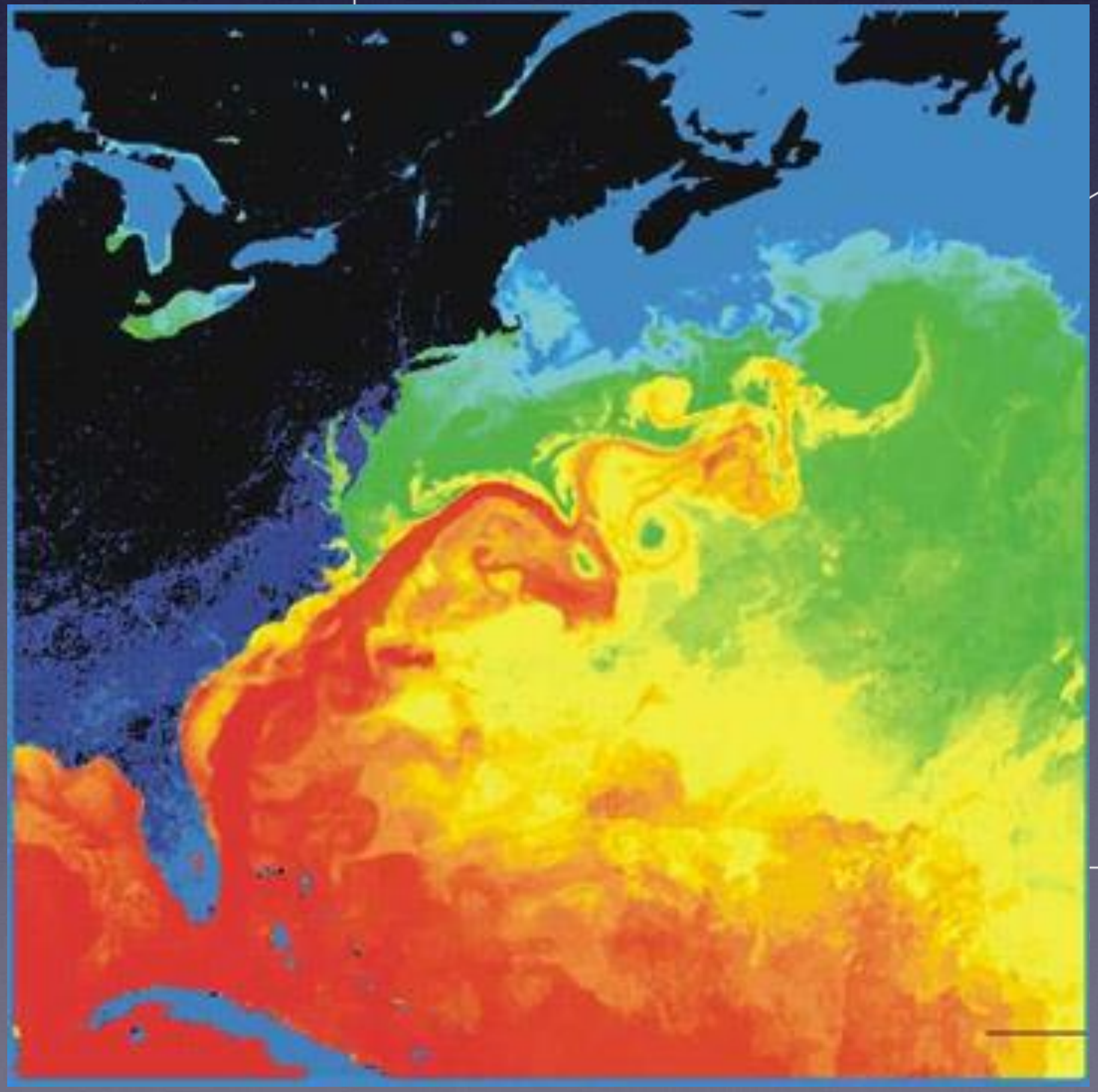


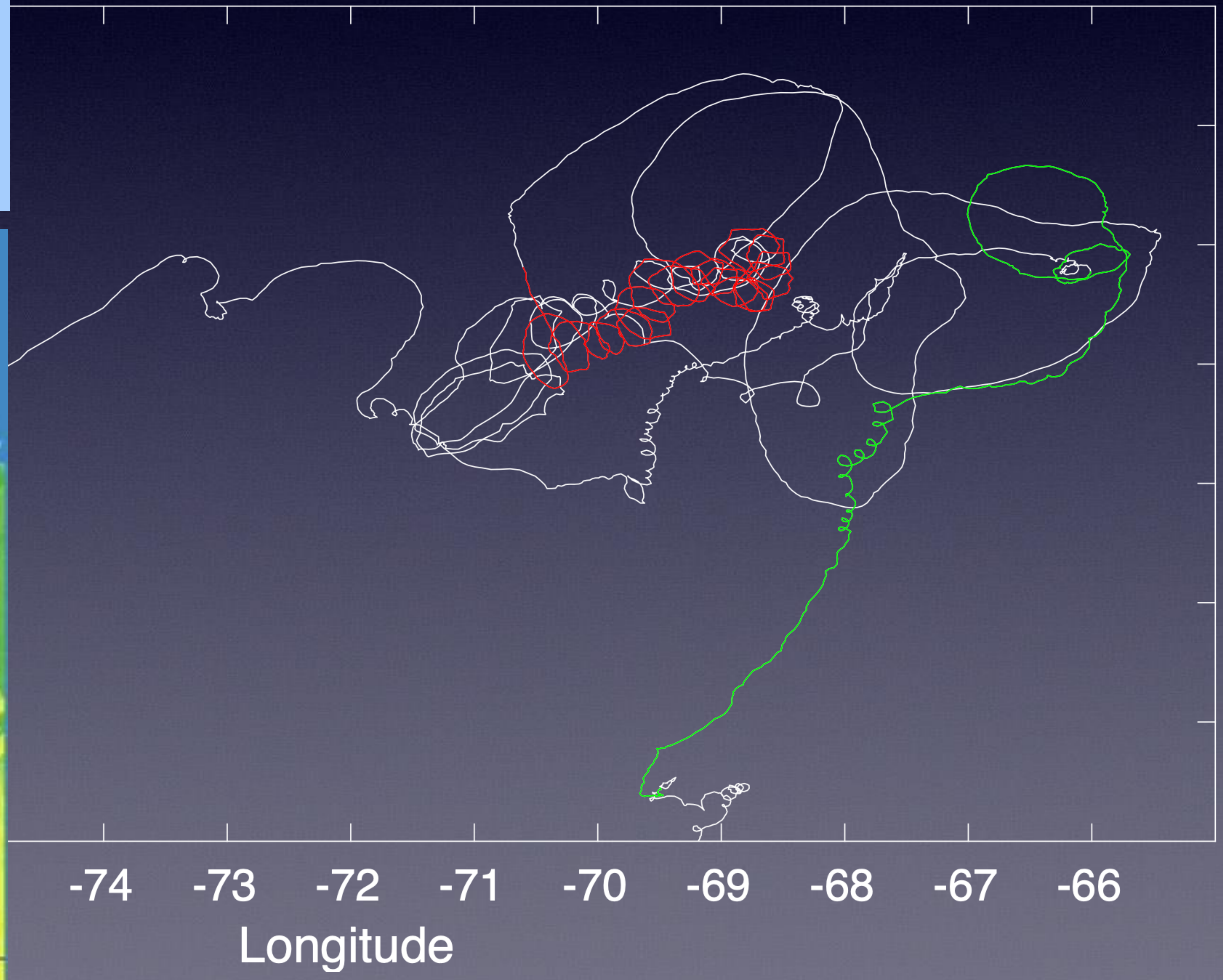
tapering



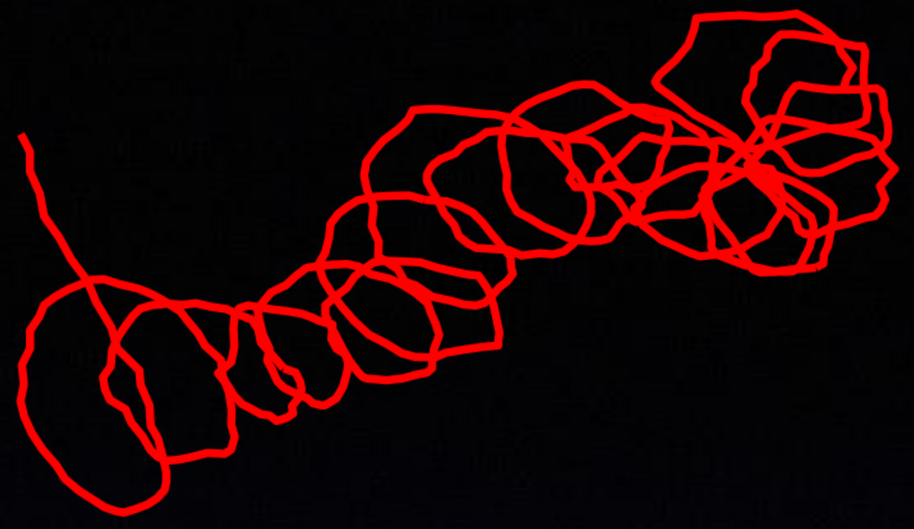
Oceanographic Drifter Data

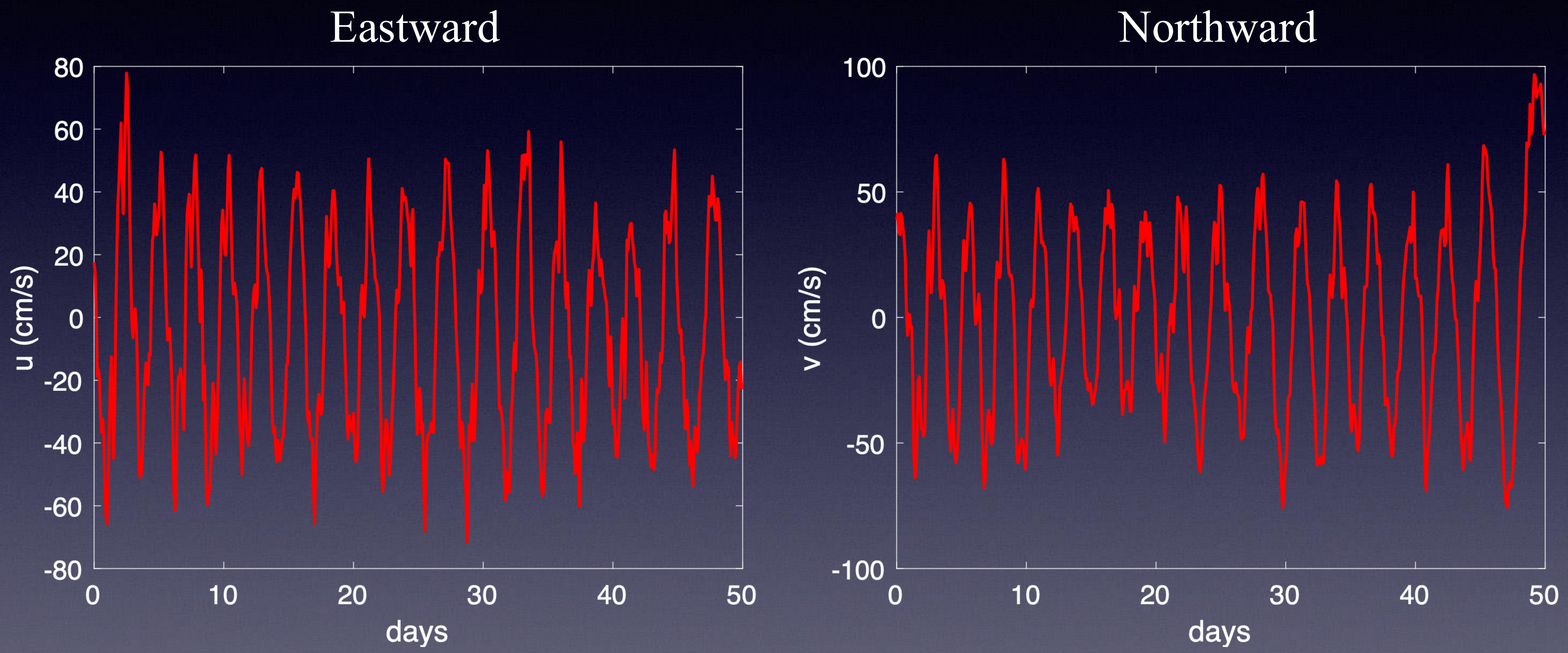






Drifter velocities for:

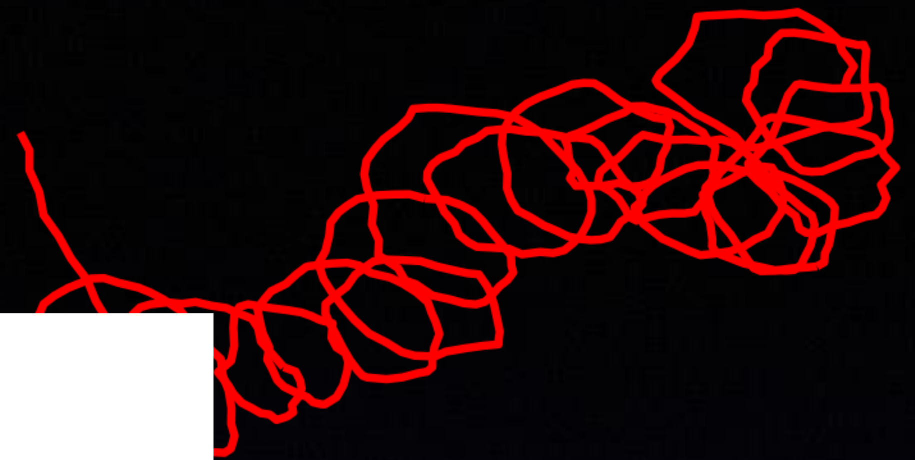


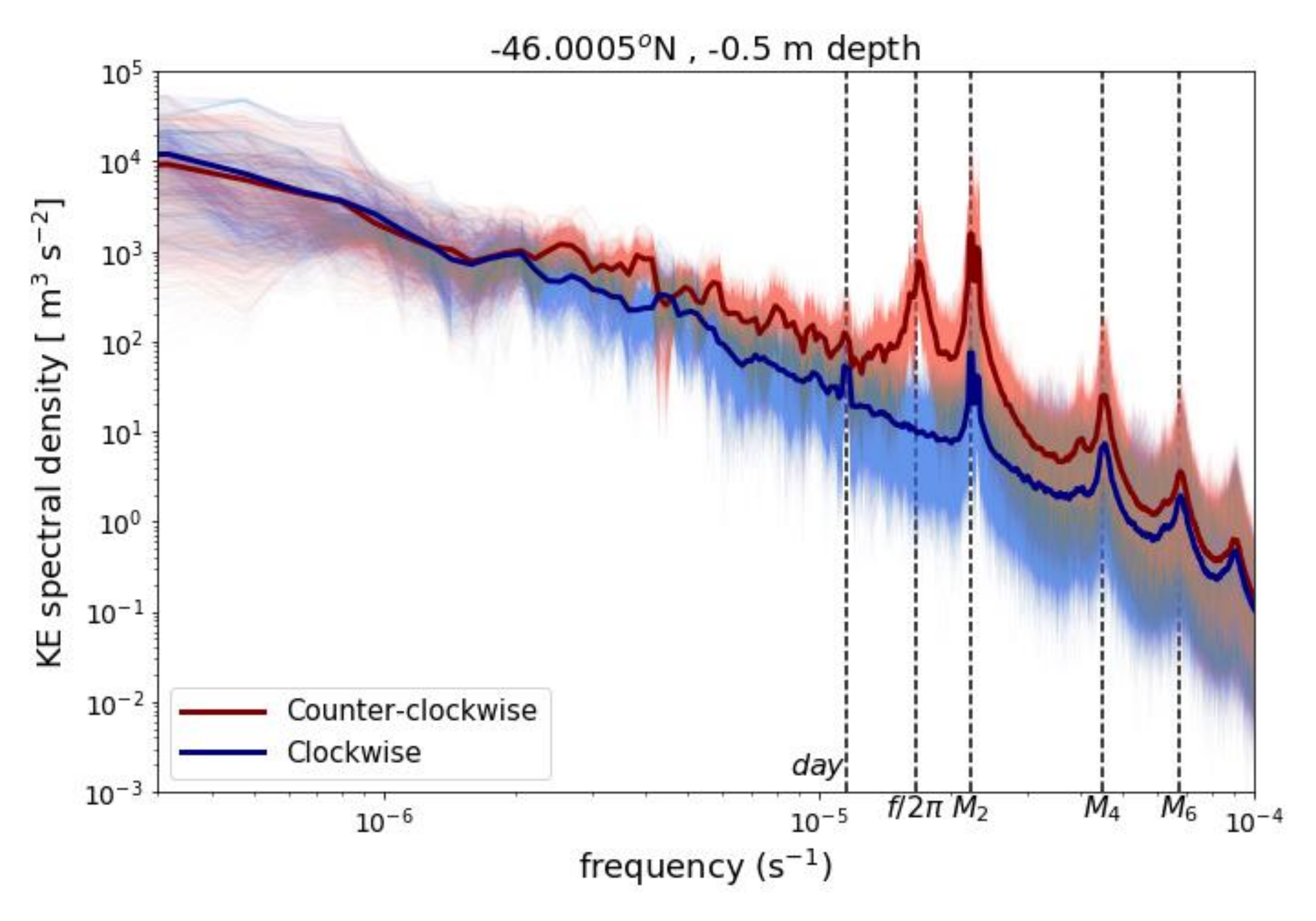


Represent particle velocities as complex-valued time series:

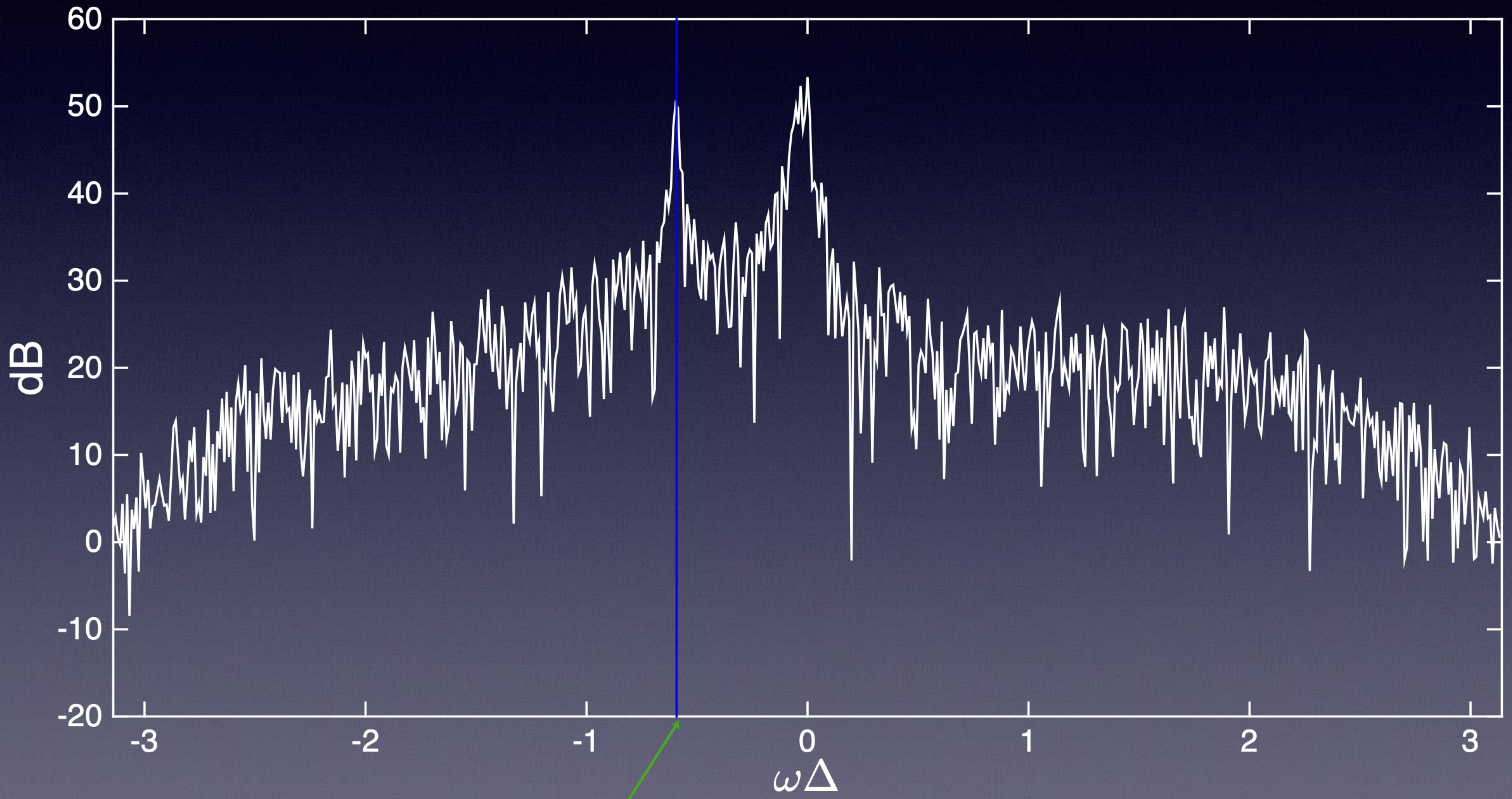
$$z_t = u_t + iv_t$$

Power spectral density for velocities from:



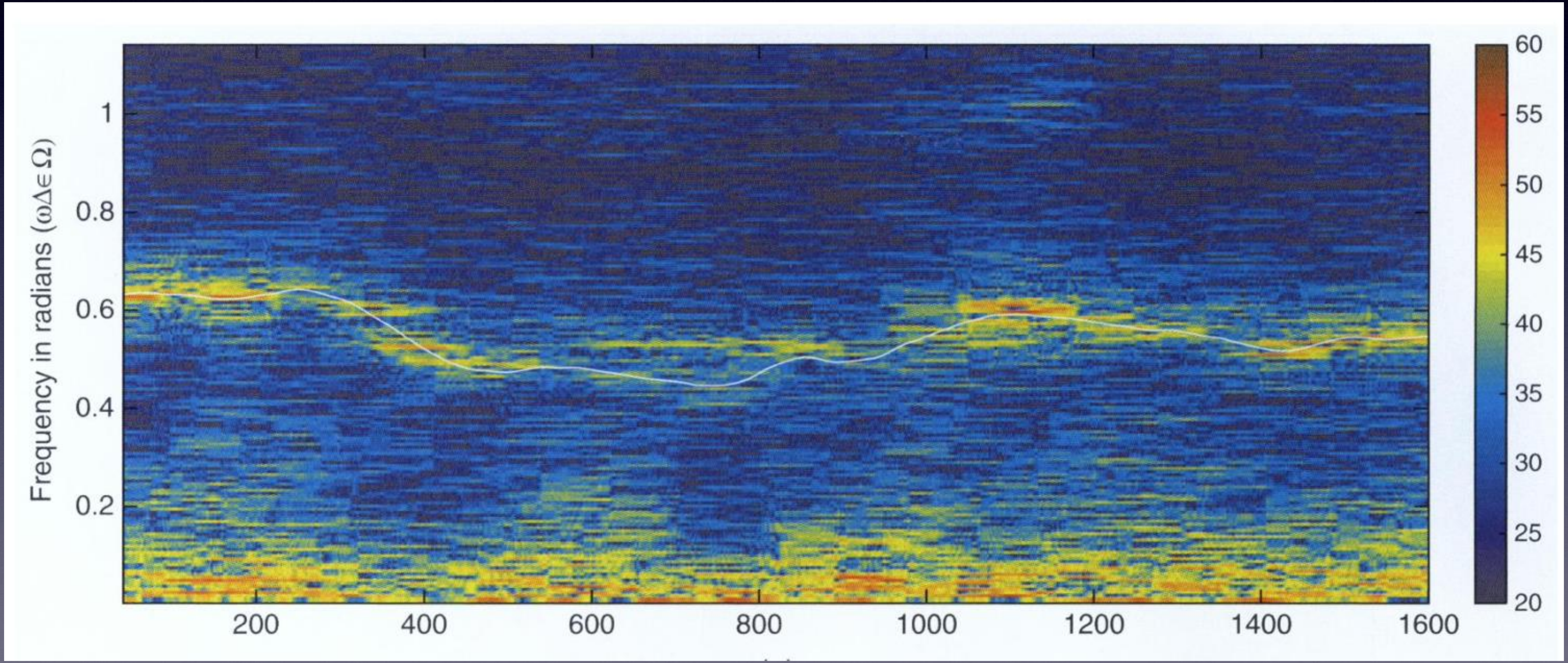


Power spectral density for velocities from:



Coriolis frequency

Nonstationarity: Time-frequency “spectrograms” and Heisenberg-Gabor uncertainty



Time in days

Peaks in the power spectral density…

Head over to RStudio for Activity 2!